

Isomorphism of nonlocal sets of orthogonal product states in bipartite quantum systems

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Highlights

- We propose two new concepts, i.e., orthogonal relation matrix, and isomorphism of different sets of OPSs.
- Two sets of OPSs have the same local distinguishability if their orthogonal relation matrices is equivalent.
- The orthogonal relation matrix of a nonlocal set of OPSs can be used to construct a new nonlocal set of OPSs.

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Abstract

To characterize nonlocal sets of orthogonal product states (OPSs) in a given bipartite quantum system, we propose two new concepts, i.e., orthogonal relation matrix of a set of OPSs and isomorphism of different sets of OPSs. In $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system, we find that two sets of OPSs, which are isomorphic to a nonlocal set of OPSs, are locally indistinguishable. In $\mathbb{C}^5 \otimes \mathbb{C}^5$ quantum system, we construct a new nonlocal set of OPSs according to the orthogonal relation matrix of a set of OPSs that cannot be perfectly distinguished by local operations and classical communication (LOCC). These results show a phenomenon that a set of OPSs is locally indistinguishable if its orthogonal relation matrix is equivalent to the orthogonal relation matrix of another set of OPSs that cannot be perfectly distinguished by LOCC in a given quantum system, and the orthogonal relation matrix of a nonlocal set of OPSs can be used to construct a new nonlocal set of OPSs. On the other hand, we give the geometric interpretation of nonlocal sets with isomorphism relation by orthogonal graphs (OGs). Our results, which can be used to classify nonlocal sets of bipartite OPSs, improve the theory of local distinguishability and deepen people's understanding of the structures of nonlocal sets.

Keywords

Quantum nonlocality without entanglement; Orthogonal product state; Orthogonal relation matrix; Isomorphism

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Introduction

In the field of quantum information, quantum nonlocality without entanglement (QNWE) is an important research content. In Ref.[1], Bennett et al. showed that there exists a set of orthogonal product states (OPSs) that cannot be reliably distinguished by local operations and classical communication (LOCC) in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system. And they called this phenomenon QNWE.

Inspired by Bennett et al.'s work, many interesting results[2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] were proposed in the past twenty years. Of all those results, most were about how to construct a set of OPSs that cannot be reliably distinguished by LOCC in a bipartite quantum system. Zhang et al.[13] generalized Bennett's result and gave a method to construct a set of OPSs that cannot be perfectly distinguished by LOCC in $\mathbb{C}^d \otimes \mathbb{C}^d$ quantum system for d is an odd number. Yu et al.[14] constructed a nonlocal set of OPSs with $2d - 1$ members in $\mathbb{C}^d \otimes \mathbb{C}^d$ quantum system. Zhang et al.[15] constructed a set of $4d - 4$ OPSs and proved its nonlocality. Wang et al.[16] gave a method to construct a nonlocal set of OPSs in $\mathbb{C}^m \otimes \mathbb{C}^n$ quantum system. Zhang et al.[17] constructed $3n + m - 4$ states that cannot be perfectly distinguished by LOCC in $\mathbb{C}^m \otimes \mathbb{C}^n$ quantum system. Compared with Wang et al.'s result, the sets constructed by Zhang et al.'s method have smaller members. Xu et al.[18] pointed out that a nonlocal set of OPSs is still locally indistinguishable even in a locally extended Hilbert space. Recently, Xu and Jiang[19] gave a new method to construct a nonlocal set of bipartite OPSs. All these results showed that it is interesting to research QNWE. In fact, a nonlocal set of OPSs in a given quantum system can be used to design quantum cryptographic protocols[20], [21], [22].

Up to now, many different nonlocal sets of bipartite OPSs have been constructed. A very natural problem is what is the relation between different nonlocal sets and how to classify those different nonlocal sets in a given quantum system. In this paper, we propose the concepts, orthogonal relation matrix of a set of OPSs and isomorphism of different sets of OPSs, which can be used to clarify different nonlocal sets of OPSs. We find that three sets of OPSs with equivalent orthogonal relation matrices have the same local distinguishability in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system. Meanwhile, we construct a new nonlocal set of OPSs according to the orthogonal relation matrix of a set of OPSs that cannot be perfectly distinguished by LOCC. Furthermore, we conclude that a set of OPSs is locally indistinguishable if its orthogonal relation matrix is equivalent to the orthogonal relation matrix of another set of OPSs that cannot be perfectly distinguished by LOCC in $\mathbb{C}^5 \otimes \mathbb{C}^5$ quantum system. And the orthogonal relation matrix of a nonlocal set of OPSs can be used to construct a new nonlocal set of OPSs. Moreover, we give the geometric interpretation of nonlocal sets with isomorphism by orthogonal graphs (OGs). Our results can be used to classify nonlocal sets of OPSs, and deepen people's understanding of the structures of nonlocal sets of OPSs.

The rest of the paper is arranged as follows. In Section2, some related knowledges and new concepts needed later are introduced. In Section3, we show three different sets of OPSs are all locally indistinguishable since they are isomorphic in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system. We construct a new nonlocal set of OPSs according to the orthogonal relation matrix of a known nonlocal set of OPSs in $\mathbb{C}^5 \otimes \mathbb{C}^5$ quantum system in Section4. In Section5, the geometric meanings of different nonlocal sets of isomorphism are given by OGs. And a brief conclusion is given in Section6.

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Section snippets

Definitions and examples

Definition 1

[18], [23]

If a set S of OPSs cannot be exactly discriminated by LOCC, we say S is locally indistinguishable or nonlocal. ...

Now, we give some new definitions that will be used in what follows.

Definition 2 Orthogonal Identifier

The orthogonal relation of bipartite OPSs $|\psi\rangle$ and $|\phi\rangle$ can be recorded in a two-dimensional vector (x, y) called orthogonal identifier, where $x = 1$ ($y = 1$) denotes $|\psi\rangle$ and $|\phi\rangle$ are orthogonal on first (second) subsystem, while $x = 0$ ($y = 0$) denotes $|\psi\rangle$ and $|\phi\rangle$ are not orthogonal on first (second) subsystem. ...

Definition 3 Orthogonal Relation Matrix

The orthogonal relations ...

...

Isomorphism of nonlocal sets of orthogonal product states in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system

In this section, we discuss the relations of different sets of orthogonal product states in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system. We find that three sets, which have the same orthogonal relation matrix, have the same nonlocality. This means that the sets with isomorphic relation have the same nonlocality. According to the orthogonal relation matrix $R_{(1)}$, we construct a set of OPSs (See Eq.(2)) in $\mathbb{C}^3 \otimes \mathbb{C}^3$.

$$|\delta_1\rangle = |0\rangle_A (|0\rangle + i|1\rangle)_B, |\delta_2\rangle = |0\rangle_A (|0\rangle - i|1\rangle)_B, |\delta_3\rangle = (|0\rangle + i|1\rangle)_A |2\rangle_B, |\delta_4\rangle = (|0\rangle - i|1\rangle)_A |2\rangle_B, |\delta_5\rangle =$$

...

Isomorphism of nonlocal sets of orthogonal product states in $\mathbb{C}^5 \otimes \mathbb{C}^5$ quantum system

In this section, we construct a set of OPSs according to the relation matrix of an existing nonlocal set of OPSs in $\mathbb{C}^5 \otimes \mathbb{C}^5$ quantum system. It is another function of orthogonal relation matrix to construct a nonlocal set of OPSs by orthogonal relation matrix of a known nonlocal set.

In Ref.[19], Xu et al. gave a general method to construct a nonlocal set of OPSs in arbitrary bipartite quantum system. By their method, we construct a set of OPSs that cannot be perfectly distinguished by LOCC in $\mathbb{C}^5 \otimes \mathbb{C}^5$...

Geometric significance of isomorphism of nonlocal sets of orthogonal product states

In this section, we discuss geometric significance of isomorphism of nonlocal sets of OPSs. In order to characterize the relations of different OPSs, DiVincenzo et al. used orthogonality

graph[27] to describe the structure of set of OPSs in a given bipartite Hilbert space.

Definition 9

[27]

Let $H = H_A \otimes H_B$ is a bipartite Hilbert space with $\dim H_A = \dim H_B$ and $S = \{|\psi_j\rangle \equiv |\varphi_{A_j}\rangle \otimes |\varphi_{B_j}\rangle \mid j = 1, 2, \dots, s\}$ is a set of orthogonal product states in H . We represent S as a graph $G = (V, E_A \cup E_B)$, where E_A and E_B are the sets of edges. ...

...

Conclusion

QNWE has attracted a lot of research interest and numerous related works[28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42] have been proposed. In this paper, we propose two new concepts, i.e., orthogonal relation matrix of a set of OPSs and isomorphism of different sets of OPSs. We find that two sets of OPSs, which are isomorphic to a nonlocal set in $\mathbb{C}^3 \otimes \mathbb{C}^3$ quantum system, are locally indistinguishable. This means that the sets with the same orthogonal ...

CRedit authorship contribution statement

Guang-Bao Xu: Conceptualization, Methodology. **Yan-Ying Zhu:** Conceptualization. **Dong-Huan Jiang:** Methodology, Writing – original draft. **Yu-Guang Yang:** Writing – review & editing. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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